In [1]:
```
# Behzad Ghabaei
# CS 82A - Data Science
# module7_lab2.ipynb
# Text Book Problem 7.1
# ratings.tsv
# Instructor Seno
# 10/13/2026
```

Problem 7.1: What drives a restaurant's overall rating? From ambience, food, and service ratings. The python code model.coef_ returns one slope per feature, rank them. Which factor should a struggling restaurant fix first?

Mathematically the Multiple Linear Regression has a goal of modeling the relationship of the outcome variable Y and the predictor variables of X. The y=mx+b formula shows: Overall Rating = B0 + B1(Ambience) + B2(Food) + B3(Service) The Overall rating is Y. The outcome and dependent variable we are trying to predict.

Ambience, Food, and Service are predictors and independent variables representing X3 X2 X1

B0 is the intercept or the constant. This is the value of the Overall rating when all the predictors are 0.

B1,B2,B3 are the coefficients. These have an impact. For example, B1 is the average change in the Overall Rating for every unit changed in Ambience, when Food and Service are the same.

There is also the error term that is unexplainable, such as: Overall rating = B0 + B1(Ambience) + B2(Food) + B3(Service) + error

Solving by hand:

If there is a row of restaurant data such as restaurant 15 with:

Food_67, Ambience_86, Service_78, Rating_64

Ambience(X1): 86

Food(X2): 67

Service(X3): 78

Actual Overall rating (Y): 64

if our coefficients were the following:

B0 = 0.5

B1 = 0.3(Ambience)

B2 = 0.3(Food)

B3 = 0.3(Service)

Y(rating) = 0.5 + 0.3(86)ambiance + 0.3(67)food + 0.3(78)service

0.5 + 86.3 + 20.01 + 23.4 = 130.21

Error says 130.21 - 64 = 66.21 difference

Solving by Python,

we want to find the coefficients with libraries like scikit-learn or statsmodels, to get the linear regression model.

In [2]:
```
import pandas as pd
from sklearn.linear_model import LinearRegression
df = pd.read_csv("ratings.tsv", sep="\t")
```

In [3]:
```
print(df.shape)
print(df.info())
print(df.describe())
print(df.head(15))
print(df.columns)
```

```
(15, 5)
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   restaurant  15 non-null     int64
 1   food        15 non-null     int64
 2   ambience    15 non-null     int64
 3   service     15 non-null     int64
 4   rating      15 non-null     int64
dtypes: int64(5)
memory usage: 732.0 bytes
None
       restaurant       food   ambience    service     rating
count   15.000000  15.000000  15.000000  15.000000  15.000000
mean     8.000000  73.133333  81.733333  75.133333  73.066667
std      4.472136   6.555986   9.786774   8.069932   8.319226
min      1.000000  64.000000  68.000000  56.000000  54.000000
25%      4.500000  68.500000  71.000000  73.500000  70.500000
50%      8.000000  72.000000  82.000000  78.000000  74.000000
75%     11.500000  77.500000  90.000000  78.000000  77.500000
max     15.000000  85.000000  96.000000  89.000000  85.000000
    restaurant  food  ambience  service  rating
0            1    85        82       89      78
1            2    80        90       80      85
2            3    83        86       83      85
3            4    70        96       75      72
4            5    68        80       78      75
5            6    65        70       56      54
6            7    64        68       61      62
7            8    72        95       72      73
8            9    69        70       78      70
9           10    75        80       75      77
10          11    75        70       75      74
11          12    72        90       78      76
12          13    81        72       78      80
13          14    71        91       71      71
14          15    67        86       78      64
Index(['restaurant', 'food', 'ambience', 'service', 'rating'], dtype='str')
```

Separate into predictors (X) and outcome (y) and Initialize and fit the Linear Regression model, then print the coefficients of each three columns and intercept.

In [4]:
```
X = df[['ambience', 'food', 'service']]
y = df['rating']

model = LinearRegression()
model.fit(X, y)

print("--- Linear Regression Model Results ---")
print(f"Intercept (Beta 0): {model.intercept_:.4f}")
print(f"Ambience Coefficient (Beta 1): {model.coef_[0]:.4f}")
print(f"Food Coefficient (Beta 2): {model.coef_[1]:.4f}")
print(f"Service Coefficient (Beta 3): {model.coef_[2]:.4f}")
```

```
--- Linear Regression Model Results ---
Intercept (Beta 0): -15.8713
Ambience Coefficient (Beta 1): 0.1338
Food Coefficient (Beta 2): 0.7338
Service Coefficient (Beta 3): 0.3240
```

The formula y = mx + b, used for Overall rating says that:

Overall rating = Intercept + Ambience + Food + Service

The optimal multiple linear regression model for the dataset is:

Rating = -15.8713 + 0.7338(Food) + 0.1338(Ambience) + 0.3240(Service)

The coefficients measure how much the overall rating increases for every 1-point increase in a specific category.

Rating (78) = -15.8713 + 0.7338 * 85(Food) + 0.3240 * 89(Service) + 0.1338 * 82(Ambience)

86.3 = -15.87 + 62.373 + 28.836 + 10.9716

86 - 78 = 8 point error difference.

Rating (Y) depends on the three predictors. Food (X1) Ambience(X2) Service(X3)

Y = B0 + B1 X1 + B2 X2 + B3 X3

B0 is the intercept or constant, and B1 B2,B3 are coefficients that impact the formula.

What drives a restaurant's overall rating?

The business recommendation I would make is that a struggling restaurant must prioritize food quality first, because it holds more than double the statistical influence more than service and over five times the influence more than ambience. According to the largest coefficient at 0.7338, Food quality is the most influential and weight carrying aspect of a restaurant's overall rating. Based on the linear regression model trained on this dataset, the features that impact a customer's satisfaction are shown as coefficients, and this three-feature regression model which fits the data, assists the business recommendation which follows the mathematics. The coefficients are obtained from ordinary least squares (OLS) from the linear regression algorithm. The features, Food, Ambience, and Service are matched as the slope: The Intercept slope is B0: -15.8713 , the Food slope is B1: +0.7338, and the Ambience Slope is B2: +0.1338, lastly the Service Slope is B3: +0.3240. The formula says: **Rating = -.1587 + 0.7338 (Food) + 0.1338 (Ambience) + 0.3240 (Service).** The Ranked Order is **Food as the Largest**

Coefficient (most influential), Service as the Median Coefficient, and Ambience as the Smallest Coefficient. In linear regression, the size of the coefficient represents the sensitivity of the outcome variable to a 1-unit change in that specific predictor. Since Food (0.7338) is the highest, a struggling restaurant will get the maximum possible increase in overall ratings by improving the restaurant kitchen and food. For example, raising the the Food score by 10 points gets a large point increase in the overall rating. However, a 10-point improvement in Ambience gets a small point increase. Thus, focusing on the food served from the kitchen, results in an efficient use of investments.

Behzad Ghabaei - End Of Lab